

NIT KURUKSHETRA, MID-SEMESTER EXAMT NATTONS-I

Course No.: CSIC 221 Course Title: Machine Learning & Data Analytics Max. Marks: 20 Duration: 50

Note The question paper contains 5 questions. All questions are compulsory.

- Q1. A researcher wants to test whether type of diet is associated with weight loss outcome using the Chi-Square test of independence at 5% significance. Data from 40 participants: Diet A—15 lost, 5 did not; Diet B—12 lost, 8 did not; Diet C—20 lost, 10 did not. State null and alternative hypotheses, compute expected frequencies, calculate the Chi-Square statistic, and conclude if diet type and weight loss outcome are independent associated. [4 Marks]
- Q2. Given dataset (2,2), (4,4), (5,5), (6,6), (8,8), (9,9), (0,4), (1,3), apply K-Means with $k=3$ using Euclidean distance. Initial clusters: $C1=(2,2),(4,4),(6,6)$, $C2=(0,4),(1,3)$, $C3=(5,5),(9,9),(8,8)$. After one iteration determine the new clusters and centroids. [5 Marks]
- Q3. Consider the following dataset: $A=\{11,21,21,22,12,10,28,45,96,12,16,98,14,49,81,15,43,92,80,96,23\}$. Calculate the Q3, Mean for the above Dataset and Create a box-plot for above data. [4 Marks]

Q4. Given the similarity matrix (Table 1), perform hierarchical clustering. Identify the two resulting clusters and construct the corresponding dendrogram. [4 Marks]

	O1	O2	O3	O4	O5	O6	O7	O8	O9	O10
O1	0	2	4	5	7	6	8	5	7	0
O2		0	3	4	6	5	7	4	6	5
O3			0	2	5	4	6	3	5	4
O4				0	4	3	5	2	4	3
O5					0	2	3	4	2	3
O6						0	4	3	3	2
O7							0	5	4	3
O8								0	2	3
O9									0	1
O10										0

Q5. Given testing data in Table 2, calculate accuracy and F-measure for each classifier, and identify which classifier has the highest F-measure. [4 Marks]

C-1: if $F1+F2+F3+F4 \geq 2$ then $C=1$ else $C=0$

C-2: if $F3+F4 > 1$ then $C=1$ else $C=0$

	Ts-1	Ts-2	Ts-3	Ts-4	Ts-5	Ts-6	Ts-7	Ts-8	Ts-9	Ts-10
F1	1	0	1	0	1	1	1	0	0	1
F2	1	1	0	0	0	1	1	1	0	0
F3	1	0	0	0	0	0	1	0	0	0
F4	0	1	1	0	0	1	1	1	0	0
Class	1	0	1	0	0	1	1	0	1	1